(iii)

$$\overset{0}{Cu} + H\overset{+5}{N}O_3 \longrightarrow \overset{+2}{Cu}(NO_3)_2 + \overset{+4}{N}o_2 + H_2O$$

$2e^-$ $1e^-$

$$\mathbf{Cu + 2HNO_3} \longrightarrow Cu(NO_3)_2 + NO_2 + H_2O$$

$$Cu + 2HNO_3 + \mathbf{2HNO_3} \longrightarrow Cu(NO_3)_2 + \mathbf{2NO_2} + 2H_2O$$

$$Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

(iv)

$$K\overset{+7}{M}nO_4 + H_2\overset{+3}{C_2}O_4 + H_2SO_4 \longrightarrow K_2SO_4 + \overset{+2}{M}nSO_4 + \overset{+4}{C}O_2 + H_2O$$

$5e^-$ $1e^- \times 2$

$$\mathbf{2KMnO_4 + 5H_2C_2O_4} + H_2SO_4 \longrightarrow K_2SO_4 + MnSO_4 + CO_2 + H_2O$$

$$2KMnO_4 + 5H_2C_2O_4 + H_2SO_4 \longrightarrow K_2SO_4 + \mathbf{2MnSO_4 + 10CO_2} + H_2O$$

$$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 10CO_2 + 8H_2O$$

45. (i) **அரை வினைகள்**

$$\overset{+7}{M}nO_4^- \longrightarrow Mn^{2+} \quad \text{..... (1)}$$

and $\quad Sn^{2+} \longrightarrow Sn^{4+} \quad \text{..... (2)}$

(1) $\Rightarrow$ $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

(2) $\Rightarrow$ $Sn^{2+} \longrightarrow Sn^{4+} + 2e^-$

(1) $\times 2 \Rightarrow$ $2MnO_4^- + 16H^+ + \cancel{10e^-} \longrightarrow 2Mn^{2+} + 8H_2O$

(2) $\times 5 \Rightarrow$ $5Sn^{2+} \longrightarrow 5Sn^{4+} + \cancel{10e^-}$

$$\Rightarrow 2MnO_4^- + 5Sn^{2+} + 16H^+ \longrightarrow 2Mn^{2+} + 5Sn^{4+} + 8H_2O$$



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